Day – 8 : Switch and Router Configuration

1 & 2. VLAN Setup and Port Assignment :

Assigning ports to VLANs isolates broadcast domains on a switch. Using interface range speeds up configuring multiple ports at once.

CLI Configuration:

Plaintext

```
Switch> enable
Switch# configure terminal

! Create VLAN 10
Switch(config)# vlan 10
Switch(config-vlan)# name Sales
Switch(config-vlan)# exit

! Create VLAN 20
Switch(config)# vlan 20
Switch(config-vlan)# name Support
Switch(config-vlan)# exit

! Assign ports FastEthernet 0/1 to 0/3 to Sales
Switch(config)# interface range FastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
```

Switch(config-if-range)# exit

! Assign ports FastEthernet 0/4 to 0/5 to Support

Switch(config)# interface range FastEthernet 0/4 - 5

Switch(config-if-range)# switchport mode access

Switch(config-if-range)# switchport access vlan 20

Switch(config-if-range)# end

Verification: To check your configuration, run:

Plaintext

Switch# show vlan brief

Verification Check: Look at the output table to verify that ports Fa0/1 - 3 are listed under 10 Sales and Fa0/4 - 5 are listed under 20 Support

3. Router-to-Router Connection and Static Route :

To connect two networks across different routers, assign IP addresses to the connecting interfaces and define a static route so Router 1 knows how to reach Router 2's local network.

Router 1 Configuration:

Plaintext

Router1> enable

Router1# configure terminal

Router1(config)# interface GigabitEthernet 0/0/0

Router1(config-if)# ip address 10.0.0.1 255.255.255.252

Router1(config-if)# no shutdown

Router1(config-if)# exit

! Static Route to reach LAN behind Router 2 (192.168.2.0/24)

Router1(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2

Router1(config)# end

Router 2 Configuration:

Plaintext

Router2> enable

Router2# configure terminal

Router2(config)# interface GigabitEthernet 0/0/0

Router2(config-if)# ip address 10.0.0.2 255.255.255.252

Router2(config-if)# no shutdown

Router2(config-if)# end

Verification Check: Run show ip route on Router 1. You should see an entry starting with S for 192.168.2.0/24. Run ping 10.0.0.2 from Router 1 to confirm physical connectivity.

4. Inter-VLAN Ping Failure & Solution :

Why the Ping Fails

When a device in the **Sales** VLAN (VLAN 10) pings a device in the **Support** VLAN (VLAN 20), the traffic drops at the switch because:

- **Layer 2 Isolation:** A switch keeps different VLANs completely separated, acting like two physical switches that aren't plugged into each other.

- **No Layer 3 Routing:** Standard switches only look at MAC addresses (Layer 2) and cannot forward traffic between separate IP subnets without a router or Layer 3 device.

How to Fix It (Solutions)

You need a device capable of **Layer 3 Routing** to bridge the traffic between the VLANs:

1. **Router-on-a-Stick (ROAS):**
   - Connect **one trunk port** on the switch to a **router port**.
   - On the router, split that single physical port into virtual **sub-interfaces** (e.g., g0/0.10 for VLAN 10 and g0/0.20 for VLAN 20).
   - Assign an IP address to each sub-interface to act as the default gateway for each VLAN.
2. **Layer 3 Switch (SVI Method):**
   - Use a Layer 3 switch, enable IP routing (ip routing), and create a virtual interface for each VLAN (interface vlan 10 and interface vlan 20).
   - Assign gateway IPs directly to those virtual interfaces on the switch.